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Symbolic Logic Framework for Situational Awareness in Mixed Autonomy

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D1.2 First version of the framework and tool integration

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Authors: Arabinda Ghosh, Sadegh Soudjani

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Project Members:

Max-Planck-Gesellschaft Zur Forderung Der Wissenschaften EV (MPI-SWS)

Technische Universiteit Eindhoven (TUE)

Kungliga Tekniska Hoegskolan (KTH)

Uppsala Universitet (UU)

Stichting Koninklijk Nederlands Lucht - En Ruimtevaartcentrum (NLR)

Siemens Industry Software Netherlands BV (SISW)

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1. Introduction

1.1 Deliverable Description

The objective of this deliverable is to develop and deliver the first version of the framework and tool integration as part of SymAware. During the first yearly consortium meeting on June 21, 2023 at Eindhoven University of Technology in Eindhoven, the Netherlands, the consortium members successfully reached a consensus and finalized the software architecture for SymAware. This architectural framework signifies a notable achievement within the project, as it establishes the basis for connecting various work packages (WPs) together. Each individual WP will be created by the respective lead beneficiaries in strong cooperation with other members of the project team. Adhering to the established framework, the initial iteration of the tool has also been formulated and made publicly accessible via the Gitlab repository (Deliverable D7.7 Software tools on public Git repository first release). The integration framework enables the smooth operations across different WPs. The anticipated outcome entails a comprehensive grasp of the project's advancement, obstacles, and possibilities, all facilitated by the harmonious interplay among diverse software elements.

1.2 Relation to Other SymAware Deliverables

In SymAware, the primary technical work packages are,

- WP1: Situational awareness (Lead: MPI-SWS)
- WP2: Spatiotemporal goal awareness (Lead: KTH)
- WP3: Risk awareness (Lead: TUE)
- WP4: Knowledge awareness (Lead: MPI-SWS)
- WP5: Use cases (Lead: NLR)
- WP6: Trustworthy and ethical human-agent interaction (Lead: UU)

Each WP contributes a unique perspective and expertise to the project, with their collective impact far greater than their individual contributions. This deliverable holds a pivotal role in unifying these technical WPs. The integration of these WPs is imperative for establishing a comprehensive and coherent project ecosystem, where the outputs of one package inform and enrich the others. By interlinking these technical work packages, the project not only achieves a holistic understanding of the *conceptual framework for awareness in multi-agent systems (MASs)* but also promotes

cross-disciplinary collaboration, enabling a more robust and innovative approach to addressing the project's overarching goals.

2. First Version of the Framework

2.1 Representation of Complex Requirements using Logic in MASs

Multi-agent systems have found successful applications in various research fields, including robotics, distributed control, resource allocation, collaborative decision support systems, and data mining, to name a few. Essentially, a multi-agent system can be envisioned as a community of decision-making entities, referred to as agents, which collaborate by sharing information within a common environment and engaging in negotiations to pursue their individual objectives or overarching collective objectives. However, the orchestration of multiple autonomous agents to achieve collaborative and coherent behavior is a formidable challenge. Within this dynamic and interconnected framework, agents must contend with a multitude of *complex requirements* that span a spectrum of considerations, from adherence to intricate traffic rules to responsiveness to ever-changing environmental conditions, and the precise pursuit of mission-specific goals. The successful coordination of these multifaceted requirements is not only essential for the safety and efficiency of MAS but also for the realization of their intended objectives, be it in the domains of autonomous transportation, robotics, or intelligent infrastructure.

This complexity motivates the employment of formal logic as a foundational tool for modeling, reasoning, and decision-making within MAS. The logic-based approach offers a structured and systematic means to navigate the intricate web of requirements, providing a solid foundation for agent behavior that ensures compliance with traffic norms, adaptability to dynamic and uncertain environmental changes, and alignment with overarching mission objectives.

Temporal logics [1] offer a structured framework for articulating temporal characteristics, providing a versatile language for specifying these properties. Linear Temporal Logic (LTL) is a formal logic used for reasoning about the temporal behavior of systems. It helps to express properties about the sequence of states in a system over time. LTL comprises a set of propositional variables, along with operators from propositional logic and temporal modalities. The propositional logic operators encompass Negation ($\neg$), Conjunction ($\wedge$), Disjunction ($\vee$), Implication ($\Rightarrow$), and Equivalence ($\Leftrightarrow$). The temporal operators encompass $\bigcirc$ (next), $\bigcup$ (until), $\Diamond$ (eventually), and $\Box$ (always). These logical and temporal operators enable us to represent complex relationships and inter-dependencies among various properties, providing us with the means to specify properties that evolve over time and depict the system's progression using LTL formulas. An LTL formula is constructed inductively according to the grammar

$$\varphi ::= \mathit{true} \mid a \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \vee \varphi_2 \mid \neg\varphi \mid \bigcirc \varphi \mid \varphi_1 \bigcup \varphi_2,\ a \in \mathcal{P}, \tag{2.1}$$

where $\mathcal{P}$ is the set of propositional variables. The LTL formula φ which satisfies a specific path σ (i.e. $\sigma \vDash \varphi$) can be defined as:

$\sigma \vDash true$,
$\sigma \vDash a$ iff $a \in \mathcal{P}$,
$\sigma \vDash \varphi_1 \wedge \varphi_2$ iff $\sigma \vDash \varphi_1$ and $\sigma \vDash \varphi_2$,
$\sigma \vDash \varphi_1 \vee \varphi_2$ iff $\sigma \vDash \varphi_1$ or $\sigma \vDash \varphi_2$,
$\sigma \vDash \neg\varphi$ iff $\sigma \nvDash \varphi$,
$\sigma \vDash \bigcirc\varphi$ iff $\sigma[1:] \vDash \varphi$ where $\sigma[i:] = s_i s_{i+1} \ldots$,
$\sigma \vDash \varphi_1 \bigcup \varphi_2$, iff $\exists i \geq 0.\sigma[i] \vDash \varphi_2$ and $\forall j < i.\sigma[j] \vDash \varphi_1$.
The other logical operators $\Rightarrow, \Leftrightarrow$ can be defined using the logical operators $\wedge, \vee, \neg$:
$\varphi_1 \Rightarrow \varphi_2 \equiv \neg\varphi_1 \vee \varphi_2$
$\varphi_1 \Leftrightarrow \varphi_2 \equiv (\varphi_1 \Rightarrow \varphi_2) \wedge (\varphi_2 \Rightarrow \varphi_1)$
$false \equiv \neg true$
The other temporal operators $\Diamond, \Box$ can be defined as:
$\Diamond\varphi \equiv true \bigcup \varphi$
$\Box\varphi \equiv \neg\Diamond\neg\varphi$

In order to demonstrate an example of the use of LTL in autonomous driving user case, let us consider a collision avoidance scenario between two vehicles (or with any obstacle Obs) i and j. The vehicles should maintain a safe distance (d_{safe}) among them or from any obstacle. This can be represented as:

$$\text{No Collision: } \Box(dist(Loc_i, Loc_j) \geq d_{safe} \wedge (dist(Loc_i, Obs) \geq d_{safe}). \tag{2.2}$$

LTL can be also used to express the environmental assumptions or to communicate the goal to the MASs. Equation 2.3 demonstrates the representation of an example where the assumption on the environment is that all the intersections will be clear. Equation (2.4) translates a simple goal like visit a location B after visiting location A:

$$\text{All intersections are } \textit{clear} \text{ infinitely often} : \Box\Diamond(\text{intersection} = \text{empty}) \tag{2.3}$$

$$\text{Visit location B after location A} : \Diamond A \wedge (A \rightarrow \Diamond B). \tag{2.4}$$

Signal Temporal Logic (STL) [2] is a predicate-oriented temporal logic designed for interpreting dense-time, real-valued signals. It offers a rigorously mathematical means of expressing tasks and regulations concerning continuous signals, all with explicit time-related semantics. STL, in particular, enables the imposition of quantitative temporal requirements, encompassing a wide range of conditions such as safety. The syntax of STL is defined as

$$\varphi ::= true \ |f(\overrightarrow{x}) > 0|\neg\varphi| \ \varphi \wedge \varphi \mid \varphi \ \mathcal{U}_I \ \varphi, \tag{2.5}$$

where $\overrightarrow{x} : \ D \rightarrow \mathbb{R}^l$ is a signal, $f : \mathbb{R}^l \rightarrow \mathbb{R}$ is a real-valued function, and $I \subseteq D$ is an interval of non negative real numbers in the time domain D. To simplify notations, we use untimed symbols like $\mathcal{U}$ to represent their timed counterparts, such as $\mathcal{U}_{[0,\infty)}$. When expressing intervals in STL formulas, we often use arithmetic expressions; for example, $\mathcal{U}_{=a}$ to denote $\mathcal{U}_a$ and $\mathcal{U}_{\leq a}$ to denote $\mathcal{U}_{[0,a]}$. Additionally, we can define various common Boolean and temporal operators using syntactic abbreviations for convenience. e.g.

$$\varphi \vee \varphi' \equiv \neg(\neg\varphi \wedge \neg\varphi'), \Diamond_I\varphi \equiv \ true \ \mathcal{U}_I\varphi, \Box_I\varphi \equiv \neg\Diamond_I\neg\varphi, \varphi\mathcal{R}_I\varphi' \equiv \neg((\neg\varphi)\mathcal{U}_I(\neg\varphi')). \tag{2.6}$$

In order to demonstrate an example of the use of STL in autonomous driving user case, let us consider the speed of a vehicle (v) should always be less than 30 km/h, and the steering angle (θ) should always be less than 20° for the first 30 s. This can be represented in the following way

$$\Box_{[0,30]}(v < 30 \wedge \phi < 20). \tag{2.7}$$

Within this project, we will use both STL and LTL to express complex requirement of the system and environment to ensure safety and resilience in the MASs framework.

2.2 Conceptual Framework for Awareness in MAS

The Conceptual Framework for Awareness in MAS presents a structured and theoretical model that helps in comprehending the notion of "awareness" within the context of MASs. In MAS, awareness refers to the agents' ability to perceive, understand, and keep track of relevant information about their environment, other agents, and the overall system state. This framework provides a systematic way to conceptualize and analyze the various components and aspects of awareness in MAS. Let us consider, a MAS with $k = 1, \ldots, n$ agents having states $s_{t,k}$ at time t as

$$s_{t,k} = \begin{pmatrix} i_{t,k} \\ x_{t,k} \\ \theta_{t,k} \end{pmatrix}. \tag{2.8}$$

In Equation (2.8), $i_{t,k}$ represents an unique id of the agent, $x_{t,k}$ represents the continuous-valued state component of the agent k i.e. position or velocity of a autonomous car or an UAV, $\theta_{t,k}$ represent the mode of agent k i.e. flight phase of an aircraft. The variable $s_{t,0} = (Env, x_{t,0}, \theta_{t,0})$ describes the states and modes of the environment e.g. location of obstacles, weather conditions.

The agents in an environment can have situational awareness about each other. The situational awareness of agent k at time t is represented as

$$\sigma_{t,k} = (\sigma^1_{t,k}, \sigma^2_{t,k}, \ldots, \sigma^n_{t,k}). \tag{2.9}$$

The situational awareness of agent k of agent j at time t consists of state awareness $\zeta^j_{t,k}$, intent awareness $\upsilon^j_{t,k}$ such that

$$\sigma^j_{t,k} = \{\zeta^j_{t,k}, \upsilon^j_{t,k}\}. \tag{2.10}$$

The variable $\zeta^j_{t,k}$ represents the information gathered by agent k regarding the current state components of agent j such that

$$\zeta^j_{t,k} = \begin{pmatrix} \hat{i}^j_{t,k} \\ \hat{x}^j_{t,k} \\ \hat{\theta}^j_{t,k} \end{pmatrix}. \tag{2.11}$$

In Equation (2.11), $\hat{i}^j_{t,k}$ represents the situational awareness of the agent k at time t regarding the identity of the agent j, $\hat{x}^j_{t,k}$ denotes the situational awareness of the agent k at time t regarding the continuous-valued state components of agent j, and $\hat{\theta}^j_{t,k}$ denotes the situational awareness of the agent k at time t regarding the discrete-values state component (modes) of agent j.

The intent awareness $\upsilon^j_{t,k}$ represents the expectations by agent k about future states of agent j. It is defined as an ordered sequence of the following component

$$\upsilon^j_{t,k} = \left((\upsilon^j_{t,k})_1, \ldots, (\upsilon^j_{t,k})_r, \ldots, (\upsilon^j_{t,k})_R \right), \tag{2.12}$$

where $(\upsilon^j_{t,k})_r$ is the expectation of agent k at time t regarding the states of agent j at a future time τ_r with $\tau_1 < \ldots < \tau_r < \ldots < \tau_R$. It is defined as

$$(\upsilon^j_{t,k})_r = \begin{pmatrix} \overline{\theta}^j_{t,k} \\ \overline{x}^j_{t,k} \\ \overline{t}^j_{t,k} \end{pmatrix}_r, \tag{2.13}$$

where $(\overline{\theta}^j_{t,k})_r$ represents the mode of agent j anticipated by the agent k, $(\overline{x}^j_{t,k})_r$ represents the continuous-valued state component of agent j anticipated by agent k, and $(\overline{t}^j_{t,k})_r$ represents the expectation of agent k at time t, concerning the time at which $(\overline{x}^j_{t,k})_r$ will be attained or the expectation of the time up to which $(\overline{\theta}^j_{t,k})_r$ will be attained by agent j.

2.2.1 Uncertainty and Risk in Situational Awareness

Risk awareness pertains to an agent's real-time comprehension of evolving risks within a multi-agent system's operations. These risks encompass the potential severity of forthcoming undesired situations and the probability of their occurrence. Therefore, an agent's grasp of risk awareness entails its perception of the likelihood of future undesirable situations. To establish this understanding, we expand the situational awareness model to include uncertainty awareness, which encompasses an agent's awareness of uncertainty related to its perception of the system's state and intent.

Uncertainty Awareness in State Awareness

The state awareness $\zeta^j_{t,k}$ in Equation (2.11) consists of awareness related to identity, continuous state variables and modes of other agents. These components of an agent can be subjected to uncertainty. The uncertainty awareness of agent k regarding the identity of agent j is represented by a set of N_I possible identities in combination with probability estimates of these identities:

$$\hat{\imath}^j_{t,k} = \begin{pmatrix} \hat{\imath}^{j,1}_{t,k} & \ldots & \hat{\imath}^{j,N_I}_{t,k} \\ \\ \hat{p}^{I,j,1}_{t,k} & \ldots & \hat{p}^{I,j,N_I}_{t,k} \end{pmatrix} \tag{2.14}$$

considering $\sum_q \hat{p}^{I,j,q}_{t,k} = 1$. For example, a pilot may think that the call–sign of some nearby aircraft is likely $\hat{p}^{I,j,1}_{t,k} = 0.9$, however in reality $\hat{p}^{I,j,2}_{t,k} = 0.1$.

Agent k's perception of uncertainty regarding a continuous state attributed to agent j is expressed through the lower and upper limits of a 95% confidence interval such that

$$\hat{x}^j_{t,k} = \left(\hat{x}^{j,L}_{t,k}, \hat{x}^{j,U}_{t,k} \right). \tag{2.15}$$

As an example, a system that measures altitude could acknowledge the range of uncertainty within its measurements, or a spectator might approximate a autonomous vehicle's speed is in the range of 30 to 50 km/h.

Agent k's perception of uncertainty concerning a mode associated with agent j is depicted as a set of N_θ potential modes coupled with probability assessments for each of these modes:

$$\hat{\theta}^{j}_{t,k} = \begin{pmatrix} \hat{\theta}^{j,1}_{t,k} & \dots & \hat{\theta}^{j,N_\theta}_{t,k} \\ \\ \hat{p}^{\theta,j,1}_{t,k} & \dots & \hat{p}^{\theta,j,N_\theta}_{t,k} \end{pmatrix} \tag{2.16}$$

considering $\sum_q \hat{p}^{\theta,j,q}_{t,k} = 1$. For example, an air traffic controller may doubt whether some an aircraft has initiated a take-off run $\hat{p}^{\theta,j,1}_{t,k} = 0.5$ or the aircraft is taxiing $\hat{p}^{\theta,j,2}_{t,k} = 0.5$.

Uncertainty Awareness in Intent Awareness

As shown in Equation (2.12), the concept of intent awareness is portrayed as a structured sequence of expected future states. This sequence includes future modes, future continuous-valued states, and projected time frames. Uncertainty can be a part of each of these components.

Agent k's acknowledgment of uncertainty concerning a future mode, denoted as $\left(\bar{\theta}^{j}_{t,k}\right)_r$, is depicted as an array of N_θ potential future modes, accompanied by probability assessments for each of these prospective modes:

$$\left(\bar{\theta}^{j}_{t,k}\right)_r = \begin{pmatrix} \bar{\theta}^{j,1}_{t,k} & \dots & \bar{\theta}^{j,N_\theta}_{t,k} \\ \\ \bar{p}^{\bar{\theta},j,1}_{t,k} & \dots & \bar{p}^{\bar{\theta},j,N_\theta}_{t,k} \end{pmatrix}_r . \tag{2.17}$$

For example, a pilot may anticipate that a nearby aircraft will likely make a right turn, but might $\left(\bar{p}^{\bar{\theta},j,1}_{t,k} = 0.9\right)$ but in reality it make a left turn $\left(\bar{p}^{\bar{\theta},j,1}_{t,k} = 0.1\right)$.

Agent k's perception of uncertainty regarding a future continuous-valued state, indicated as $\left(\bar{x}^{j}_{t,k}\right)_r$, is represented through the lower and upper limits of a 95% confidence interval:

$$\left(\bar{x}^{j}_{t,k}\right)_r = \left(\bar{x}^{j,L}_{t,k}, \bar{x}^{j,U}_{t,k}\right)_r . \tag{2.18}$$

As an example, an air traffic controller anticipates that an aircraft will be moving at a speed within the range of 25 to 35 km/h.

Agent k's awareness of uncertainty regarding the anticipated time, denoted as $\left(\bar{t}^{j}_{t,k}\right)_r$, for achieving a future mode or state value is conveyed through the lower and upper limits of a 95% confidence interval:

$$\left(\bar{t}^{j}_{t,k}\right)_r = \left(\bar{t}^{j,L}_{t,k}, \bar{t}^{j,U}_{t,k}\right)_r . \tag{2.19}$$

For example, an air traffic controller could anticipate that an aircraft will commence its take-off run within a specific time window, or a driver might foresee that a car will reach a speed of 100 km/h in a time frame spanning from 8 to 12 seconds.

Risk Awareness

An agent k's awareness of risk signifies the agent's comprehension of the probability of unfavorable future situations occurring during their operations. The awareness of risks of agent k' associated with agent i at time t is denoted by a number of risks denoted by $\left(\rho^j_{t,k}\right)_r$:

$$\left(\rho^j_{t,k}\right)_r = \begin{pmatrix} \beta^j_{t,k} \\ \eta^j_{t,k} \\ p^j_{t,k} \\ \alpha^j_{t,k} \\ I^j_{t,k} \end{pmatrix}_r . \tag{2.20}$$

In Equation (2.20), $\left(\beta^j_{t,k}\right)_r$ signifies a category of risks, such as the possibility of an aircraft collision, collision with an object, collision involving a person, collision with the ground, or running out of battery. It can also encompass non-safety-related risks, like the chance that airplane passengers miss their connecting flights or the possibility of a package not being delivered on schedule. The term $\left(\eta^j_{t,k}\right)_r$ signifies one or more severity classes of the possible future condition, e.g. catastrophe (major, minor, negligible). The term $\left(p^j_{t,k}\right)_r$ signifies probability estimates of the severity classes of the possible future condition, which can be a value or a probability class, e.g. almost certain, likely, moderate, unlikely. $\left(\alpha^j_{t,k}\right)_r$ denotes the acceptability of the risk, e.g. acceptable, aarginal, unacceptable. Finally, the term $\left(I^j_{t,k}\right)_r$ signifies a set of identities of agents that are involved in the risky situation, e.g. another aircraft that poses a collision risk. The set is empty if there are no other agents involved in the particular risky condition.

The risk awareness $\rho^j_{t,k}$ is added as a component to the situational awareness $\sigma^j_{t,k}$. Risk awareness for agent k may be absent when considering other agents, denoted as j. For example, a drone operator might only focus on the risks associated with the specific drone they are piloting. The number of agents for which an agent maintains non-empty risk awareness depends on their role and responsibilities within the operation. For instance, an air traffic controller is accountable for a group of aircraft within a sector and must uphold situational awareness for all of these aircraft.

2.3 The Proposed Framework (First Version)

Drawing inspiration from the existing framework of awareness from section 2.2 (for further details please refer to Deliverable 5.1), within this project we have developed the first framework for awareness in MAS as shown in Fig. 2.1. In addition to the previously mentioned aspect of situational awareness, our framework includes the integration of knowledge, perception, communication, planning, control, the physical state of the system, and communication interfaces with human agents. A brief summary of each component is provided below for reference.

- **Knowledge**: The knowledge base helps in informed decision making by storing and managing relevant information related to coordination, collaborative efforts, learning, and adaptive behaviors [3]. For instance, in an autonomous driving scenario, the knowledge base can

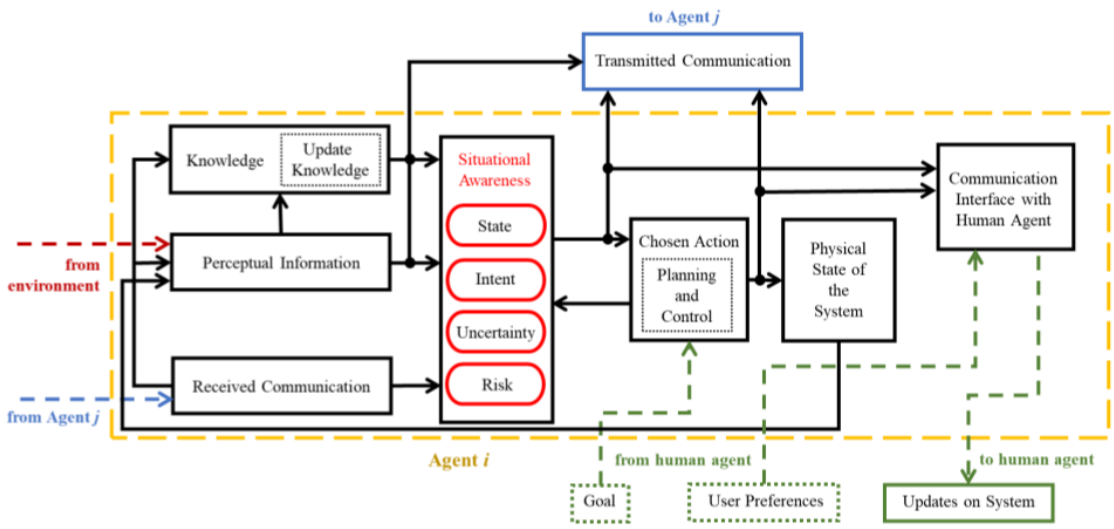

Figure 2.1: First version of the framework in SymAware.

contain information on road layouts, intersections, lanes, traffic rules, speed limits, and traffic signs.

- **Estimation and perception**: The role of estimation and perception is to provide agents with an understanding of their environment, enabling them to make informed decisions and effectively interact with their surroundings [4]. This involves utilizing various sensing mechanisms to gather data from the environment. The collected data is then processed and interpreted using intelligent algorithms, such as signal processing techniques and object recognition, to transform raw sensory input into meaningful representations. For example, in a driving scenario, the agent can use cameras, lidar, radar, GPS to detect the presence of other vehicles, pedestrians, or obstacles.

- **Communication**: The primary role of the communication module is to facilitate effective communication and coordination among the agents within the system. This involves transmitting and receiving information that is essential for collaborative tasks, decision-making, and synchronized actions [5].

- **Situational awareness**: Situational awareness involves gathering and processing information about the surroundings, including the state and intent of other agents, uncertainty of the environment, and risk (referred to Deliverable D5.1). By maintaining situational awareness, agents can make informed decisions and adapt their behaviors based on real-time information. This awareness enables agents to anticipate changes, respond effectively to dynamic situations, and collaborate more efficiently with other agents in the system. Situational awareness is a fundamental aspect of intelligent decision-making, as it empowers agents to align their actions with the current state of the environment, contributing to the overall success of the

MAS by enabling agents to act intelligently and purposefully within their context.

- **Planning and control**: The role of planning and control is to enable agents to effectively strategize, sequence actions, and regulate their behaviors to achieve specific goals and objectives. The planning aspect involves generating a sequence of actions that will lead the agent toward its desired outcomes. Control, on the other hand, involves executing the planned actions while continuously monitoring the environment for changes.

- **Physical state of the system**: The physical state of the system refers to the tangible attributes and conditions of the system. In the context of an autonomous driving scenario, the physical state refers to the dynamics of the vehicle (the physics-based model). The system can be modeled in discrete time as $M(\theta)$ parameterized with $\theta = [\theta_1, \ldots, \theta_N]$ and divided into N subsystems such that $M^j(\theta^j),\ j \in 1, \ldots, N$ as

$$M^j(\theta^j) : \begin{cases} x^j(k+1) = & f^j(x^j(k), u^j(k), \theta^j) + w^j(k) \\ y^j(k) = & h^j(x^j(k)) \end{cases} \tag{2.21}$$

where the state, input, and observation of the jth subsystem $M^j(\theta^j)$ at the kth time-step are denoted by $x^j(k) \in \mathbb{X}^j$, $u^j(k) \in \mathbb{U}^j$ and $y^j(t) \in \mathbb{Y}^j$, respectively [6]. The input of the system could depend on state of other subsystems thus creating coupling between them. The dynamics can also be considered in continuous time.

As an example, consider a system with N agents having a nonlinear model of an autonomous vehicle with 7 dimension. For the jth agent, variables x_1^j and x_2^j signifying the the position coordinates, x_3^j signifying the steering angle, x_4^j signifying the heading velocity, x_5^j signifying the yaw angle, x_6^j signifying the yaw rate, and x_7^j signifying the slip angle [7]. The equations of motion for $j = [1, \ldots, N]$ is given by

$$\text{for } |x_4^j(k) < 0.1| : \tag{2.22}$$

$$x_i^j(k+1) = x_i^j(k) + \tau a_i + 0.5\zeta_i(k),\ i \in \{1, \ldots, 7\} \backslash \{3, 4\}, \tag{2.23}$$

$$x_3^j(k+1) = x_3^j(k) + \tau Sat_1(u_1^j) + 0.5\zeta_3(k), \tag{2.24}$$

$$x_4^j(k+1) = x_4^j(k) + \tau Sat_2(u_2^j) + 0.5\zeta_4(k), \tag{2.25}$$

$$\text{and for } |x_4^j(k) \geq 0.1| : \tag{2.26}$$

$$x_i^j(k+1) = x_i^j(k) + \tau b_i + 0.5\zeta_i(k),\ i \in \{1, \ldots, 7\} \backslash \{3, 4\}, \tag{2.27}$$

$$x_3^j(k+1) = x_3^j(k) + \tau Sat_1(u_1^j) + 0.5\zeta_3(k), \tag{2.28}$$

$$x_4^j(k+1) = x_4^j(k) + \tau Sat_2(u_2^j) + 0.5\zeta_4(k), \tag{2.29}$$

where, variables u_1^j and u_2^j are inputs and they control the steering angle and heading velocity, respectively of the jth agent and $Sat_1(\cdot)$ and $Sat_2(\cdot)$ are saturation functions of the inputs. For further details on the system parameters, we direct the readers to the work [7].

- **Communication interface with human agents**: The communication interface with human agents serves as a bridge that facilitates effective and meaningful interaction between human users and the autonomous agents within the system. This interface will be continuously updated according to findings from studies with human participants, and it will reflect

their expressed needs and preferences about the interaction with the agents. In this way, the interface will allow the human users to convey instructions, queries, or requests to the agents; and will enable the agents to send continuous updates about their state (expressed via their Situational Awareness) and the state of the environment.

Collectively, the expected outcomes of these modules contribute to the overall success of the MAS by enabling agents to operate intelligently, communicate efficiently, adapt to changing conditions, and collaborate effectively, all within the context of their operational environment and human interactions.

3. Tool Integration

3.1 Achieved Objective

Based on the framework from Chapter 2, a tool is developed and can be accessed using the following link

https://gitlab.mpi-sws.org/sadegh/eicsymaware/

The description of the tool is available in Deliverable D7.7. The objective of the developed tool is to create a versatile and efficient platform that empowers users and agents within the MAS to collaborate seamlessly, make informed decisions, and achieve shared goals. This tool aims to facilitate the integration of various modules and components within the MAS framework, fostering enhanced communication, coordination, and situational awareness among agents.

4. Conclusion

In conclusion, the development and realization of the first version of the framework and tool integration mark a step towards the overarching goals of SymAware project in line of the Horizon EIC 2021 Pathfinder challenge. Through meticulous planning, design, and implementation, we have established a cohesive platform that harmonizes various modules within the MAS. While the current version of the code does not yet include a human component, discussions are underway among our partners to incorporate human agents into the existing code and use cases. With the release of the first version of the framework, we look forward to building upon our progress and continuing to advance towards the development of the released tool by integrating the models of human agents.

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